Online Food Delivery Analysis: Data-Driven Business Insights

Customer & Order Analysis

8.Distance vs delivery delay analysis.

Delivery time shows no relationship with distance. Deliveries over 30 km are no slower than deliveries under 5 km. About 41% of deliveries take over 150 minutes and 60% take over 60 minutes, whatever the distance.

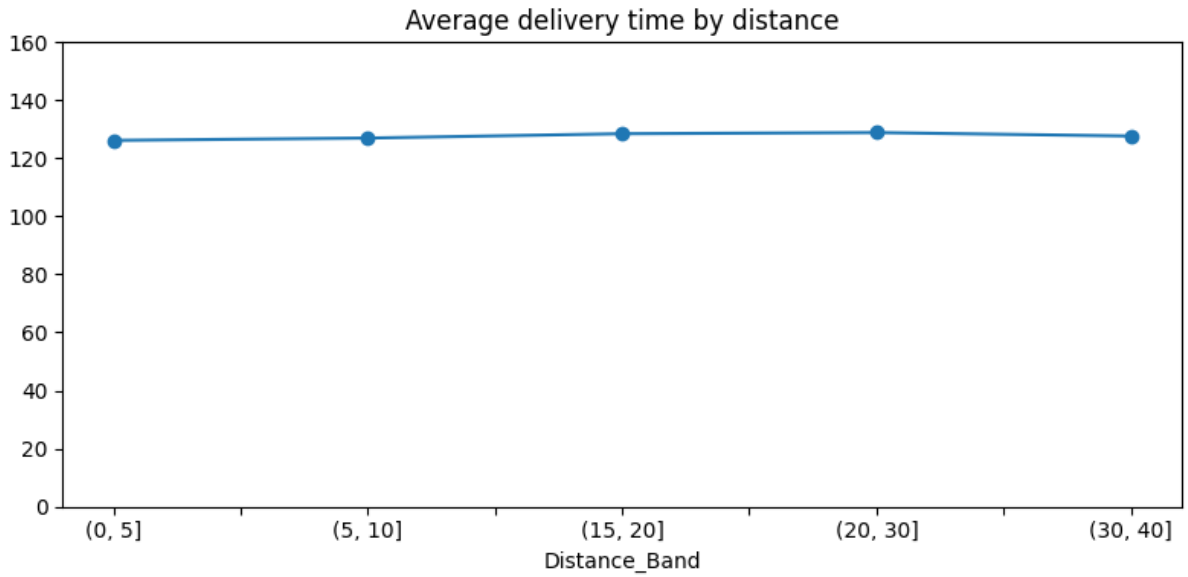
Illustrative sample records :

Distance	orders	avg_min	delayed
(0, 5]	8434	126.2	40.5
(5, 10]	10447	127.0	41.1
(10, 15]	7	140.0	42.9
(15, 20]	3805	128.5	41.9
(20, 30]	7603	128.9	42.1
(30, 40]	7490	127.7	41.8

2 . VISUALIZATIONS

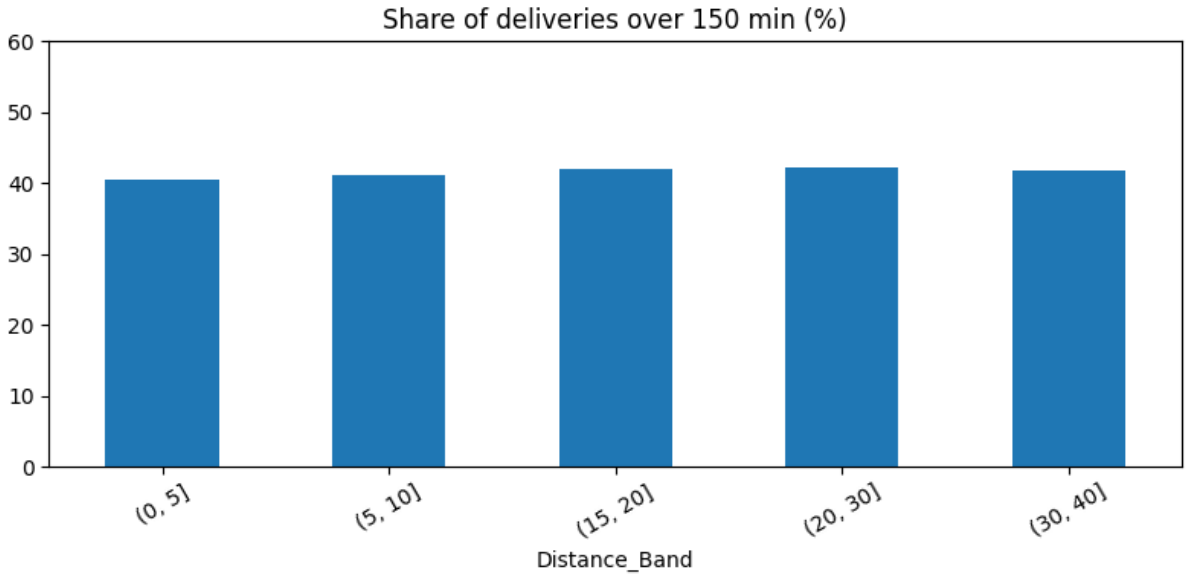
Visualization – 1

Below line chart shows average delivery time by distance band.



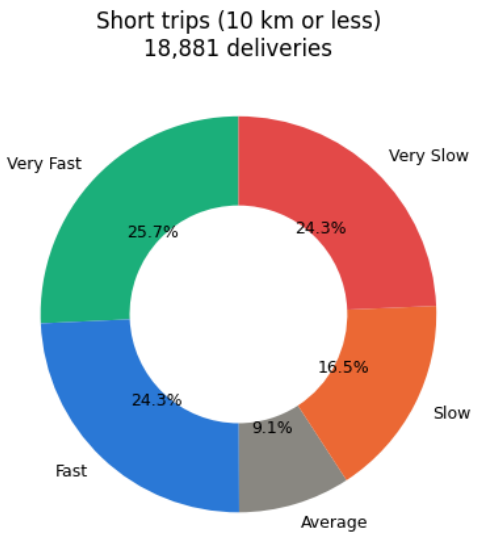
Visualization – 2

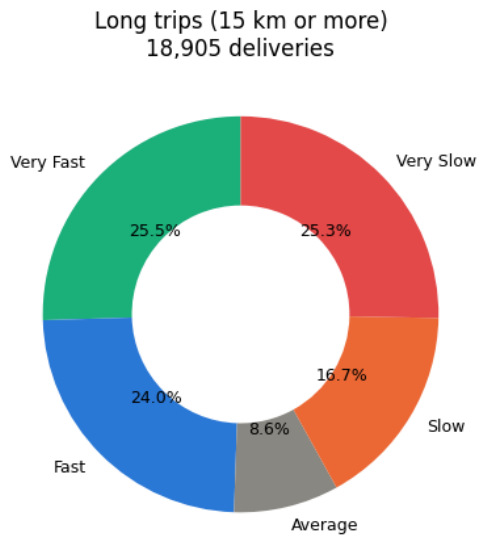
This bar chart shows the share of deliveries that took more than 150 minutes in each distance band.



Visualization – 3

The two donuts compare delivery speed for short trips (10 km or less) and long trips (15 km or more), and they look almost the same.





3.ANALYTICAL INSIGHTS – DEMAND AND REVENUE

INSIGHT – 1: Half of demand comes from close in.

50% of orders come from within 10 km and 49% from 15 to 40 km, but the close range covers only 10 km against 25 km for the far range. So demand is about 2.5 times denser near the restaurant even though the average distance is 16.4 km.

INSIGHT – 2: Delivery time doesn't grow with distance.

Average delivery time is 126 to 129 minutes in every band. Each extra kilometre adds about 3 seconds, so a 1 km trip and a 40 km trip take about the same time.

INSIGHT – 3: Profitability doesn't change either.

Average profit margin is 14- 15% and average discount ₹92 to ₹95 in every band.

INSIGHT – 4: Cancellations don't rise with distance.

The cancellation rate is 14.6% to 15.2% across bands. Customers living far from the restaurant don't cancel more.

4.ANALYTICAL INSIGHTS-OPERATIONS AND CUSTOMER EXPERIENCE:

INSIGHT-6: Establish realistic delivery-time

Establish realistic delivery-time estimates based on distance and reduce avoidable delays.

INSIGHT- 7: Estimate Distance vs. Delivery Cost

Optimize delivery zones and dispatching to control costs while maintaining service quality. Fix standard delivery cost based on the distance so that it will help customer to make order easily.